THE UNIVERSITY OF WESTERN AUSTRALIA

FINA2205
Practical Finance Analytics

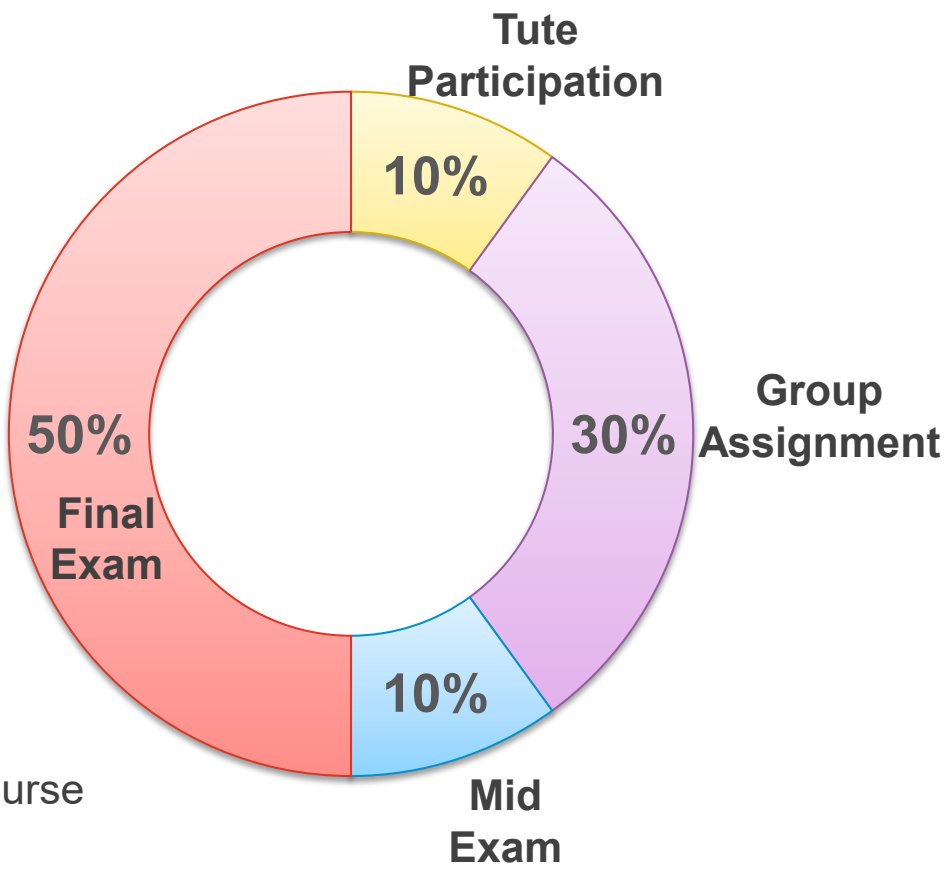
Lecture 1:
Review of Financial Statistics I

Reading: Levine et al (Chapter 3)

AACSB ACCREDITED EFMD EQUIS ACCREDITED

Coordinates

- **Dr Yihui Lan**
  - yihui.lan@uwa.edu.au
- **Lectures**
  - Thur 2am-4pm
- **Course content or Admin questions:**
  - **Discussion Board**
- **Consultation hours**
  - After lectures
- **Course Announcements**
  - LMS: https://lms.uwa.edu.au/ultra/course



The Assessments

Participation (10%)

Students are required to **prepare, attend and participate** at least 5 out of the 8 tutorials throughout the semester.

Group Assignments (30%)

Students will be asked to complete a group assignment in a group of 4-5 students.

Examinations (60%)

The mid exam (10%) will be an online MCQ exam via LMS.

The final examination (50%) will be two hours. Further details will become available later in the semester.

Please see the unit outline for due dates.

Lecture 1 Outline

I. An Overview of Quant Finance

II. Financial Returns: Basics

III. Review of Financial Statistics

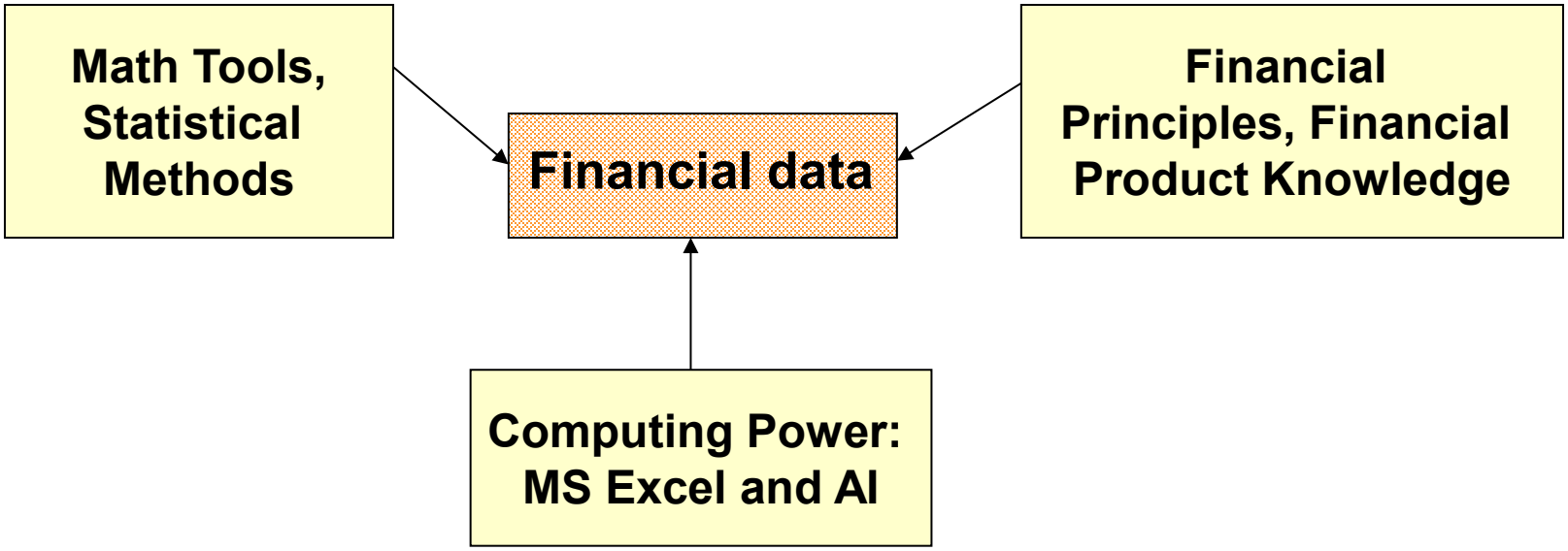
1. Central Tendency
2. Measures of Variation
3. Measures of Shape
4. Descriptive Statistics

IV. In-lecture activities

I. Quantitative Finance: What, Why, How?

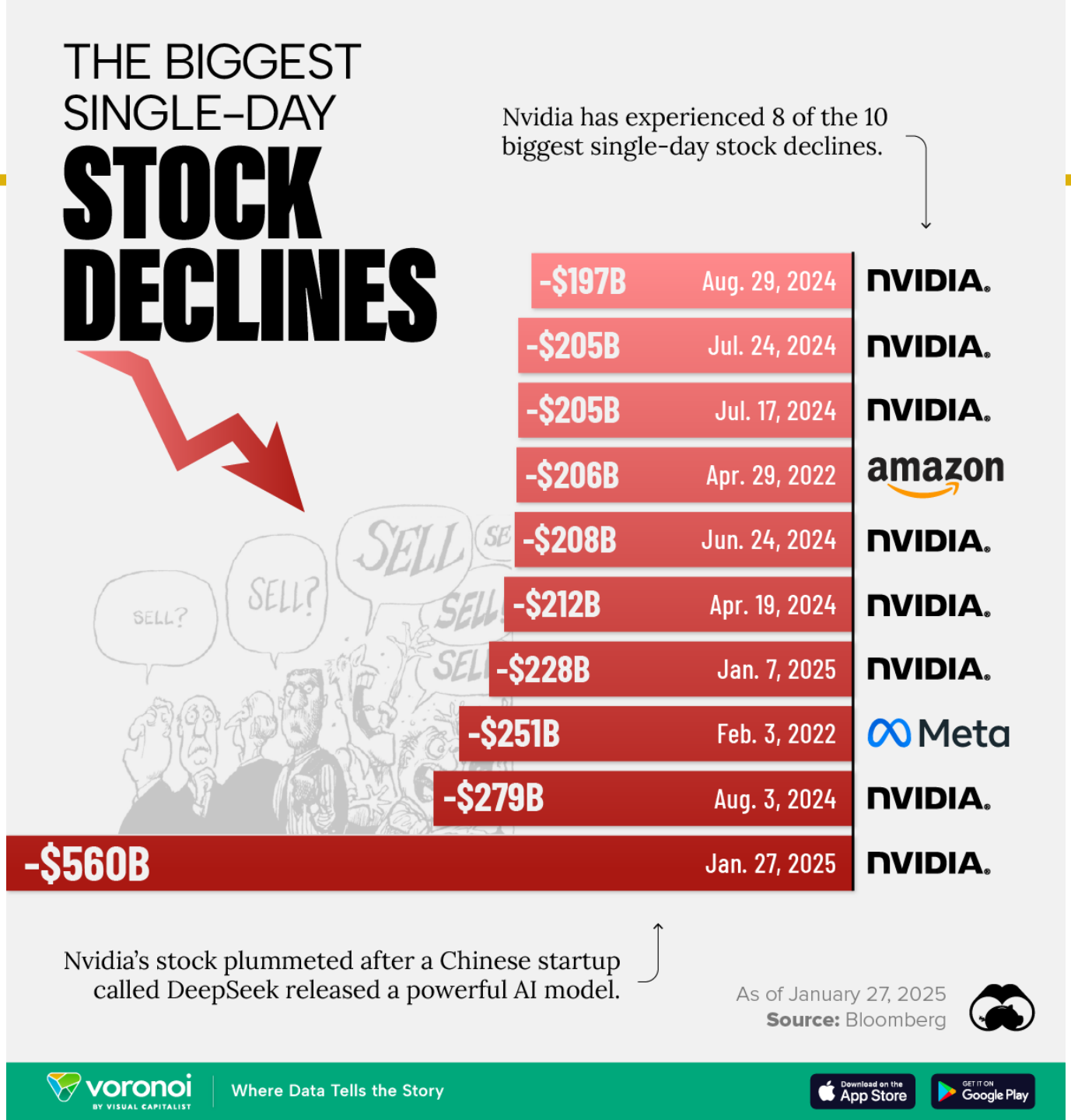
- A collection of basic topics of a broad discipline called financial engineering or quantitative finance
  - Analysis of risk and how to manage the risk

- Three areas



I. Quantitative Finance: What, Why, How?

- Phenomenal growth: (i) financial market (ii) risk management tools
- Important events impacting finance and the economy
  - GFC 2007-2009 -- the worst recession since the Great Depression (1929-32)
  - 2013: the euro area sovereign debt crisis
  - 2014: emerging markets in trouble
  - 2015: China Stock market turmoil, oil price volatility
  - 2016: energy crisis, Brexit
  - 2017: Trumponomics
  - 2018-9: U.S. trade war against China, Brexit
  - 2020: Brexit, Coronavirus
  - 2021: Covid vaccine
  - 2022: Russian invasion of Ukraine by land, air and sea in Feb
  - 2023: Rising interest rates and inflation, collapse of Silicon Valley Bank, Israel–Hamas war
  - 2024: Red Sea crisis, August Market Meltdown, Trump's successful re-election
  - 2025: DeepSeek a potential "game changer" in the AI world?, etc



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https://www.visualcapitalist.com/deepseek-tanks-stock-market-nvidia/

AI in Finance

Source: https://www.youtube.com/watch?v=zQVn0BGgx0A

I. Quantitative Finance: What, Why, How?

Quantitative methods failed to save us from financial failures.

Shall we consign Quant methods to the dustbin of history?

- "We have witnessed the death of a planet. Call it Planet Finance."
  (Source: http://www.vanityfair.com/politics/features/2008/12/banks200812)

I. Quantitative Finance: What, Why, How?

- Shall we give up medical science if there is a outbreak of epidemic disease?
- Shall we discredit all scientists such as Newton, Navier and Stokes when we witnessed the failure of Space Shuttle Challenger in 1986?

OBVIOUSLY NO !!!

- That a theory/method might not work in practice is not because it is totally flawed. It is possible that
  - not every assumption is met
  - it has not been applied properly
  - people do not know exactly how the theory works
  - etc.

I. Quantitative Finance: <u>What</u>, Why, How?

- Attending lectures and labs
- Learning strategies
  - Critical thinking: relating theories to the real world
  - Hands-on: practicing in classes, doing homework, practice questions and assignment

II. Financial Returns: Basics

Although financial analysts often observe prices on their screens such as stock prices, commodity prices, bond prices and exchange rates, the **main objective** of quantitative finance is:

to analyse
financial returns.

II. Financial Returns: Basics

Simple Returns: one period

- Let P_t be the price of an asset at time t and let P_{t-1} be the price of the asset at time t – 1.
- Assuming that the financial asset does not pay any dividends
- The one-period (for instance, one-day, one-week, one-month or one-year) simple NET return is

$$R_t = \frac{P_t - P_{t-1}}{P_{t-1}} = \frac{P_t}{P_{t-1}} - 1$$

- The one-period simple GROSS return is

$$1 + R_t = \frac{P_t}{P_{t-1}}$$

II. Financial Returns: Basics

Simple returns: an example

- Consider a one-month investment in a BHP share.
- You bought the stock in month t–1 at $90 and sold it in month t for $100.
- Calculate the simple net return and the gross return of holding the share over the one-month period.

- Solution
  - 1m simple net return $R_t = \frac{100-90}{90} = 11.11\%$
  - 1m simple gross return $1+R_t = \frac{100}{90} = 111.11\%$

II. Financial Returns: Basics

Log return

- In this unit, we will base most of our examples on continuously-compounded returns.

- The continuously compounded one-period return (logarithmic return, or simply log return) is given by

$$r_t = \ln(1+R_t) = \ln\left(\frac{P_t}{P_{t-1}}\right) = \ln P_t - \ln P_{t-1}$$

- Why is $r_t \approx R_t$, when both returns are small?
  - Mathematically, for small x, we have $x \approx \ln(1+x)$

III. Review of Financial Stats

Definitions

- The **central tendency** is the extent to which all the data values group around a typical or central value.
- The **variation** is the amount of dispersion or scattering of values
- The **shape** is the pattern of the distribution of values from the lowest value to the highest value.

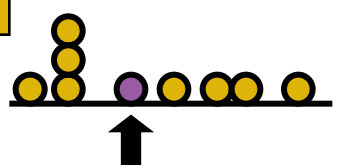
III.1 Measures of Central Tendency: Summary**

Central Tendency

Arithmetic Mean

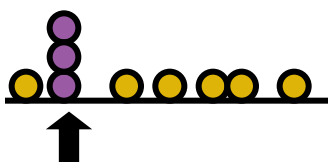
$$\overline{X} = \frac{\sum_{i=1}^{n} X_i}{n}$$

Median



Middle value in the ordered array

Mode



Most frequently observed value

Geometric Mean

$$\overline{X}_G = (X_1 \times X_2 \times \cdots \times X_n)^{1/n}$$

Rate of change of a variable over time

III.1 Central Tendency

E.g. House Prices:

	$2,000,000
	$ 500,000
	$ 300,000
	$ 100,000
	$ 100,000
Sum	$ 3,000,000

- **Mean:** ($3,000,000/5)
  = **$600,000**
- **Median:** middle value of ranked data
  = **$300,000**
- **Mode:** most frequent value
  = **$100,000**

Measures of Central Tendency: Which measures to choose?

- The **mean** is generally used, unless extreme values (outliers) exist.
- The **median** is often used, since the median is not sensitive to extreme values. For example, median home prices may be reported for a region; it is less sensitive to outliers.
- In some situations it makes sense to report both the **mean** and the **median**.

III.1 Central Tendency**

Measures of Central Tendency: Geometric Mean

- Geometric mean
  - Used to measure the rate of change of a variable over time

$$\overline{X}_G = (X_1 \times X_2 \times \cdots \times X_n)^{1/n}$$

- Financial application: geometric mean rate of return
  - Measures the status of an investment over time

$$\overline{R}_G = [(1+R_1)\times(1+R_2)\times\cdots\times(1+R_n)]^{1/n} - 1$$

  - Where R_i is the rate of return in time period i

III.1 Central Tendency**

Measures of Central Tendency: Example for Geometric Mean Rate of Return

An investment of $100,000 declined to $50,000 at the end of year one and rebounded to $100,000 at end of year two:

$$X_1 = \$100{,}000 \qquad X_2 = \$50{,}000 \qquad X_3 = \$100{,}000$$

50% decrease 100% increase

The overall two-year return is zero, since it started and ended at the same level.

III.1 Central Tendency**

Measures of Central Tendency: Example for Mean Rate of Return

Solution: Use the 1-year returns to compute the arithmetic mean and the geometric mean:

Arithmetic mean rate of return:

$$\overline{X} = \frac{(-.5)+(1)}{2} = .25 = 25\%$$

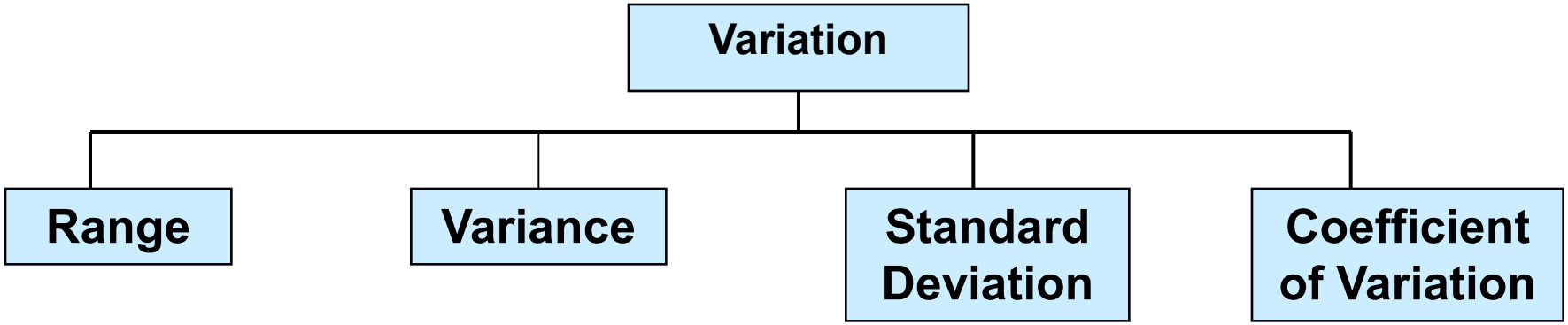
Misleading result

Geometric mean rate of return:

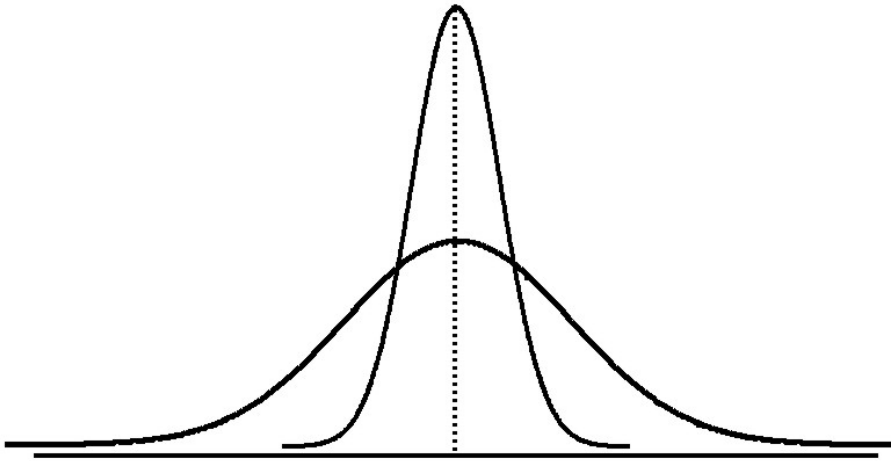
$$\begin{aligned}\overline{R}_G &= [(1+R_1)\times(1+R_2)\times\cdots\times(1+R_n)]^{1/n} - 1 \\ &= [(1+(-.5))\times(1+(1))]^{1/2} - 1 \\ &= [(.50)\times(2)]^{1/2} - 1 = 1^{1/2} - 1 = 0\%\end{aligned}$$

Sensible result

III.2 Measures of Variation**



- Measures of variation give information on the **spread** or **variability** or **dispersion** of the data values.



Same center, different variation

III.2 Measures of Variation

Measures of Variation: The Sample Variance

- Average (approximately) of squared deviations of values from the mean

  – Sample variance:

$$S^2 = \frac{\sum_{i=1}^{n}(X_i - \overline{X})^2}{n-1}$$

Where $\overline{X}$ = arithmetic mean

n = sample size

X_i = i^{th} value of the variable X

III.2 Measures of Variation

Measures of Variation: The Sample Standard Deviation

- Most commonly used measure of variation
- Shows variation about the mean
- Is the square root of the variance
- Has the same units as the original data
  - Sample standard deviation:

$$S = \sqrt{\frac{\sum_{i=1}^{n}(X_i - \overline{X})^2}{n-1}}$$

Measures of Variation: The Sample Standard Deviation

Steps for Computing Standard Deviation

1. Compute the difference between each value and the mean.
2. Square each difference.
3. Add the squared differences.
4. Divide this total by n-1 to get the sample variance.
5. Take the square root of the sample variance to get the sample standard deviation.

The Sample Standard Deviation: Example

Sample Data (X_i) : **10 12 14 15 17 18 18 24**

n = 8 Mean = $\overline{X}$ = 16

$$S = \sqrt{\frac{(10-\overline{X})^2 + (12-\overline{X})^2 + (14-\overline{X})^2 + \cdots + (24-\overline{X})^2}{n-1}}$$

$$= \sqrt{\frac{(10-16)^2 + (12-16)^2 + (14-16)^2 + \cdots + (24-16)^2}{8-1}}$$

$$= \sqrt{\frac{130}{7}} = \boxed{4.3095}$$

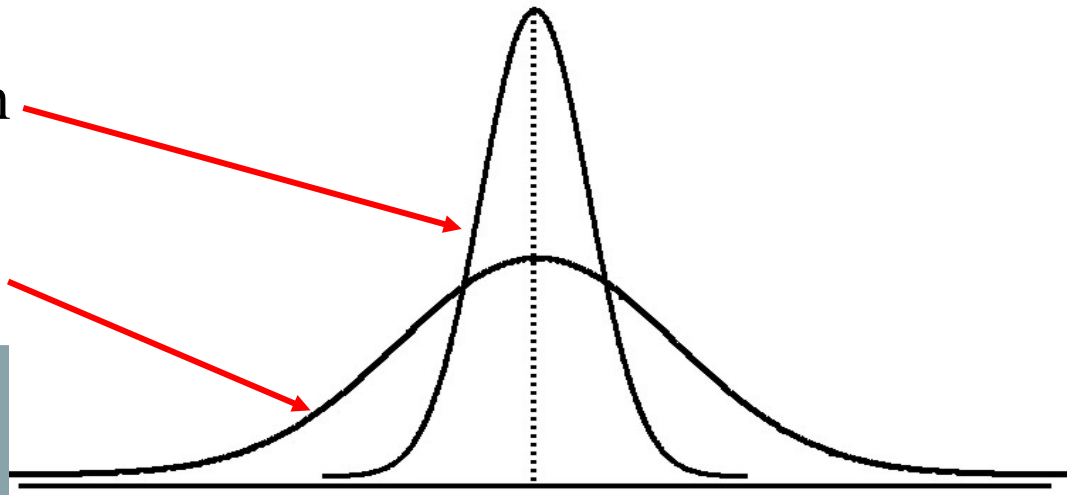
A measure of the "average" scatter around the mean

III.2 Measures of Variation

Comparing the Standard Deviation

Smaller standard deviation

Larger standard deviation



III.2 Measures of Variation

Summary

- The more the data are spread out, the greater the range, variance, and standard deviation.
- The more the data are concentrated, the smaller the range, variance, and standard deviation.
- If the values are all the same (no variation), all these measures will be zero.
- None of these measures are ever negative.

III.2 Measures of Variation

Coefficient of Variation

- Measures relative variation
- Always in percentage (%)
- Shows variation relative to mean
- Can be used to compare the variability of two or more sets of data measured in different units

$$CV = \left(\frac{S}{\overline{X}} \right) \cdot 100\%$$

III.2 Measures of Variation

Coefficient of Variation

- Stock A:
  - Average price last year = \$50
  - Standard deviation = \$5

$$CV_A = \left(\frac{S}{\overline{X}}\right) \cdot 100\% = \frac{\$5}{\$50} \cdot 100\% = 10\%$$

- Stock B:
  - Average price last year = \$100
  - Standard deviation = \$5

$$CV_B = \left(\frac{S}{\overline{X}}\right) \cdot 100\% = \frac{\$5}{\$100} \cdot 100\% = 5\%$$

Both stocks have the same standard deviation, but stock B is less variable relative to its price

III.2 Measures of Variation

Coefficient of Variation

- Stock A:
  - Average price last year = \$50
  - Standard deviation = \$5

$$CV_A = \left(\frac{S}{\overline{X}}\right) \cdot 100\% = \frac{\$5}{\$50} \cdot 100\% = 10\%$$

- Stock C:
  - Average price last year = \$8
  - Standard deviation = \$2

$$CV_C = \left(\frac{S}{\overline{X}}\right) \cdot 100\% = \frac{\$2}{\$8} \cdot 100\% = 25\%$$

Stock C has a much smaller standard deviation but a much higher coefficient of variation

Locating Extreme Outliers: Z-Score

$$Z = \frac{X - \overline{X}}{S}$$

where X represents the data value

X-bar is the sample mean

S is the sample standard deviation

- To compute the Z-**score** of a data value, subtract the mean and divide by the standard deviation.
- The Z-score is the number of standard deviations a data value is from the mean.
- A data value is considered an extreme outlier if its Z-score is less than -3.0 or greater than +3.0.
- The larger the absolute value of the Z-score, the farther the data value is from the mean.

III.2 Measures of Variation

Locating Extreme Outliers: Z-Score

- Suppose the mean math SAT score is 490, with a standard deviation of 100.
- Compute the Z-score for a test score of 620.

$$Z = \frac{X - \overline{X}}{S} = \frac{620 - 490}{100} = \frac{130}{100} = 1.3$$

A score of 620 is 1.3 standard deviations above the mean and would not be considered an outlier.

III.3 Shape of a Distribution

- Describes how data are distributed
- Two useful shape related statistics are:
  - Skewness
    - Measures the extent to which data values are not symmetrical
  - Kurtosis
    - Kurtosis affects the peakedness of the curve of the distribution—that is, how sharply the curve rises approaching the center of the distribution

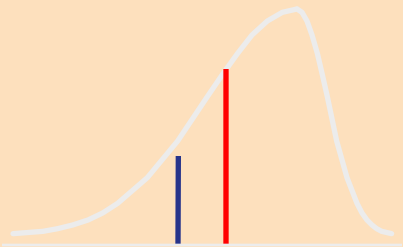
III.3 Shape of a Distribution**

Skewness

- Measures the extent to which data is not symmetrical

Left-Skewed

Mean < Median

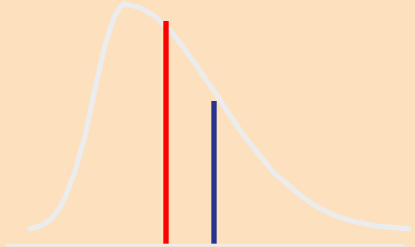


Symmetric

Mean = Median

Right-Skewed

Median < Mean

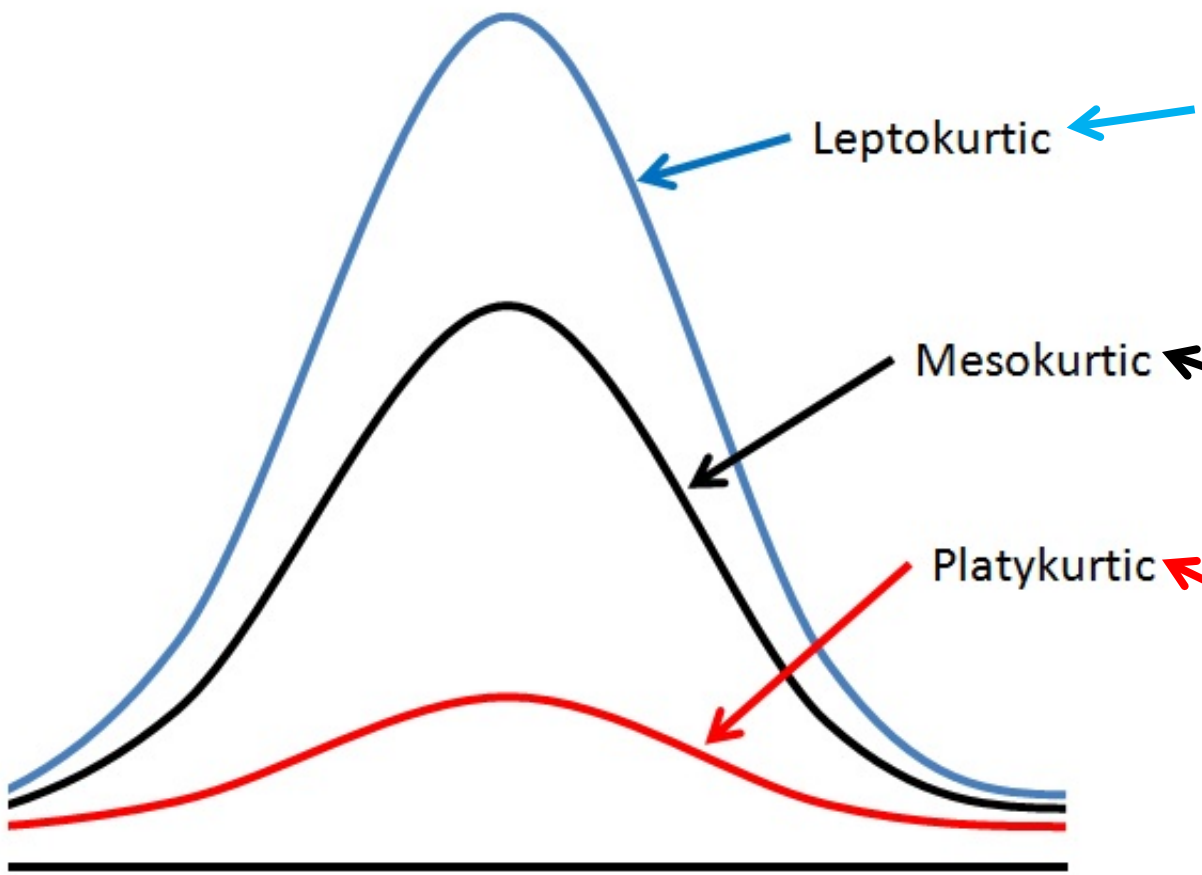


Skewness Statistic	< 0	0	>0

III.3 Shape of a Distribution**

Kurtosis

measures how sharply the curve rises approaching the center of the distribution)



Sharper Peak Than Bell-Shaped (Kurtosis > 0)

Bell-Shaped (Kurtosis = 0)

Flatter Than Bell-Shaped (Kurtosis < 0)

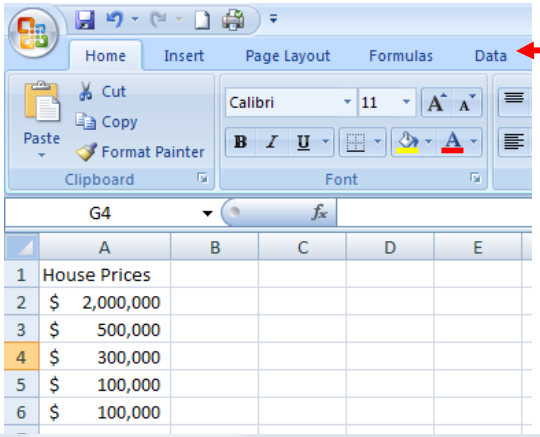
III.4 Descriptive Statistics**

Using Microsoft Excel Functions

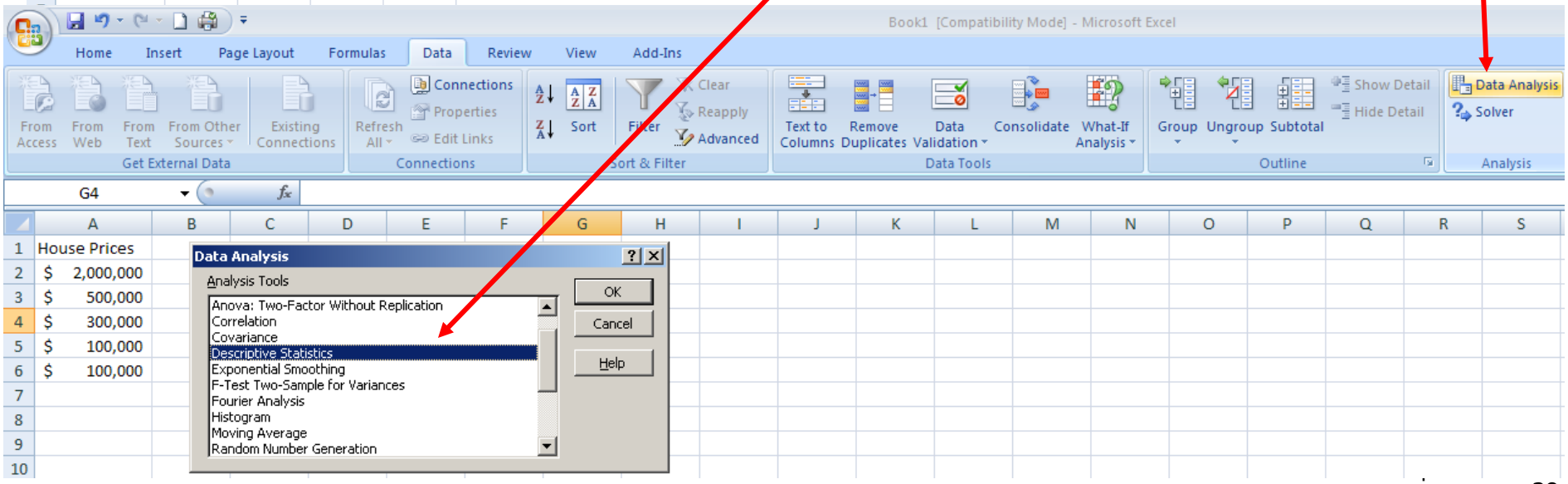
House Prices		Descriptive Statistics		
$ 2,000,000		Mean	$ 600,000	=AVERAGE(A2:A6)
$ 500,000		Standard Error	$ 357,770.88	=D6/SQRT(D14)
$ 300,000		Median	$ 300,000	=MEDIAN(A2:A6)
$ 100,000		Mode	$ 100,000.00	=MODE(A2:A6)
$ 100,000		Standard Deviation	$ 800,000	=STDEV(A2:A6)
		Sample Variance	640,000,000,000	=VAR(A2:A6)
		Kurtosis	4.1301	=KURT(A2:A6)
		Skewness	2.0068	=SKEW(A2:A6)
		Range	$ 1,900,000	=D12 - D11
		Minimum	$ 100,000	=MIN(A2:A6)
		Maximum	$ 2,000,000	=MAX(A2:A6)
		Sum	$ 3,000,000	=SUM(A2:A6)
		Count	5	=COUNT(A2:A6)

III.4 Descriptive Statistics**

Using Microsoft Excel Functions



1. Select Data.
2. Select Data Analysis.
3. Select Descriptive Statistics and click OK.



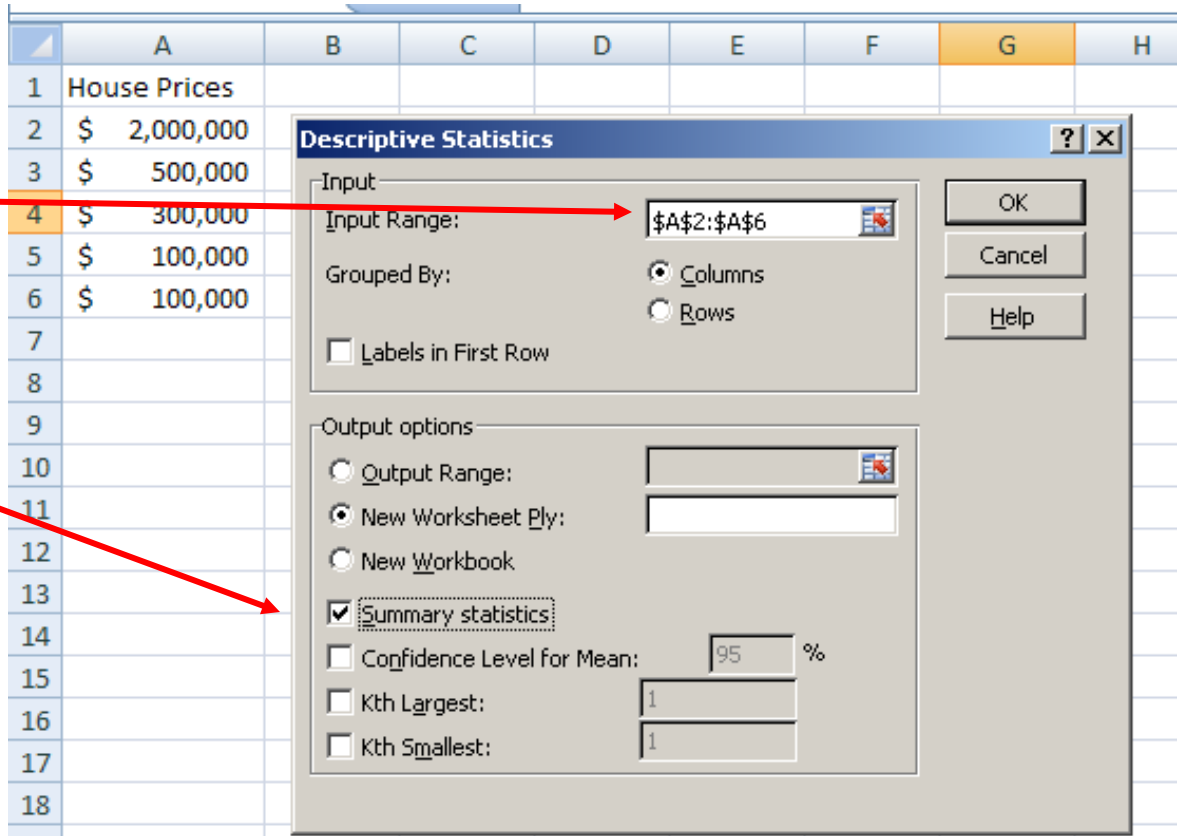
III.4 Descriptive Statistics**

Using Microsoft Excel Functions

4. Enter the cell range.

5. Check the Summary Statistics box.

6. Click OK



IV. Summary and in-lecture activities

- Returns:

  Simple net/gross returns, log(cont. compounded) return

- Stats Review

  Central Tendency: arithmetic/geometric mean, median, mode

  Measures of Variation: range, Var/Stdev, CV

  Measures of Shape: skewness, kurtosis

  Descriptive Measures: Excel functions, Excel add-in

- MS Excel